

THE META-PROMPT

That Generated a Production-Grade Enterprise System Design

Domain: Medical Devices | Order Management Automation | Microsoft 365

Prompt Architecture: Niteesh Gupta | Commercial Excellence and AI Systems

AI-ASSISTED SYSTEM DESIGN | DOMAIN-DRIVEN PROMPT ENGINEERING | ENTERPRISE ARCHITECTURE

What Is a Meta-Prompt — And Why It Works

A standard prompt asks an AI to produce an output. A Meta-Prompt engineers the AI's reasoning process before it produces anything. Instead of asking 'Build me an order management system,' this prompt forces the model to:

- Inhabit four distinct corporate personas simultaneously — each with conflicting priorities
- Identify blind spots from each stakeholder's perspective before proposing any solution
- Apply real industry constraints as hard guardrails — including precise exception routing rules
- Output a specification structured for immediate low-code implementation, not a conceptual overview

The Core Insight: When you encode your domain constraints into the prompt itself — including nuanced rules like 'credit limit breach auto-routes to Sales Org, not the dealer' — the model has no room to hallucinate a generic solution. It reasons within your actual business reality. Domain expertise is the architecture. The prompt is the delivery mechanism.

Structural Breakdown — 4 Engineering Layers

Layer 1	Role Assignment
The prompt opens by assigning a primary role (Principal Solution Architect) and a stakeholder panel of four personas. This multi-persona assignment forces the model to hold competing priorities in tension — exactly how real enterprise system design works. No single perspective dominates the output.	
Layer 2	Business Constraints as Hard Rules
Three non-negotiable rules are defined with explicit failure outcomes. Framed as HARD RULES, not requirements. Nuanced routing logic — such as credit limit rejections going to Sales Org rather than the dealer — is specified at this layer so the model encodes it as a system rule, not an optional consideration.	
Layer 3	Stakeholder Review Gate
Before the model touches technical architecture, it reviews the process from each stakeholder's perspective. This surfaces blind spots — business continuity gaps, dealer relationship impacts, audit trail requirements — that a pure technical prompt would miss entirely. The review gate is non-optional.	
Layer 4	Output Format Specification
The prompt specifies exact section names, required content per section, and detail standard (pseudocode, not placeholder logic). Tone specified as professional, authoritative, exhaustive. This prevents the model from producing a high-level summary dressed as a specification.	

Copy-Ready — Paste Directly into Any LLM

The prompt below is the exact architecture used to generate the AOGS System Design Blueprint. Works with any frontier LLM (Claude, GPT-4o, Gemini Ultra). Details in [brackets] should be replaced with your organisation's specifics.

SECTION 1 — ROLE ASSIGNMENT

System Prompt: Enterprise Solution Architect & Workflow Engineer

Role: Act as a Principal Solution Architect specialising in Microsoft 365 Enterprise integrations, working alongside a panel of expert stakeholders: a Medical Devices Supply Chain Director, a Commercial Excellence Head, a Corporate Finance Auditor, and a Low-Code Implementation Engineer.

Objective: Transform the following high-level business blueprint for an Automated Order Governance System (AOGS) into an exhaustive, production-grade functional design specification.

SECTION 2 — BUSINESS SCENARIO & HARD CONSTRAINTS

1. THE BUSINESS SCENARIO & CONSTRAINTS

A medical devices company processes orders via a shared Outlook inbox (3 operators).
Inputs: Standardised Excel order sheets or unstructured Hospital Purchase Orders (PDFs).
Senders: Dealers or Internal Sales Org.

HARD RULE 1 (Compliance): If Sales Org sends an order, the Dealer MUST be in CC.
If missing, the process must terminate immediately with automated notification.

HARD RULE 2 (Logic): Order quantities must perfectly align with Pack Sizes.
No partial packs allowed.

HARD RULE 3 (Logic): Total order value must meet the dealer-specific MOV threshold.

Approval Flow:

- Validated orders route to Finance for Credit Control check via Adaptive Card approval.
- APPROVED: System auto-generates the Global Billing Excel and emails it directly to the Global ERP team. No manual step from Supply Chain or operators.
- REJECTED – Credit Limit Breach: System auto-emails Sales Org with Finance comments. Sales Org resolves credit position with Finance and resubmits. No manual forwarding.
- REJECTED – Other Reason: System auto-emails original sender with Finance comments.

SECTION 3 — STAKEHOLDER REVIEW GATE (MANDATORY PRE-BUILD)

2. EXECUTE A CRITICAL STAKEHOLDER REVIEW

Before defining technical architecture, review from each perspective:

1. Supply Chain Director: Change management, resource shift from data entry to exception handling, and business continuity if the system goes down.
2. Commercial Excellence Head: Sales hierarchy enforcement, TAT optimisation, and impact on dealer relationships and notification tone.
3. Corporate Finance Auditor: Risk mitigation, fraud prevention via authenticated approvals, data lineage ensuring the final Excel matches validated input, and auditability of the credit limit rejection routing rule.
4. Low-Code Implementation Engineer: Power Automate connector selection, AI Builder model training requirements, Microsoft Lists schema including Finance_Rejection_Type column, and failure recovery within a single flow architecture.

SECTION 4 — OUTPUT FORMAT SPECIFICATION

3. EXPECTED OUTPUT REQUIREMENT

Generate a detailed Technical Blueprint Specification buildable immediately.
Structure into three sections:

SECTION A: Microsoft 365 Architecture Mapping

- Outlook Shared Inbox: trigger config and CC filtering logic
- Power Automate Cloud Flows: full orchestration from intake to auto-dispatch
- AI Builder Document Processing: PDF extraction strategy and confidence thresholds
- Microsoft Lists: full schema including Finance_Rejection_Type and Global_Dispatch_DateTime columns

SECTION B: Step-by-Step Logic and Exception Handling

Pseudocode/conditional logic covering all stages and exceptions:

- Exception 1: Sales Org email missing Dealer in CC
- Exception 2: Hospital PDF product code not in Master Data
- Exception 3: Order fails Pack Size or MOV check (handle separately)
- Exception 4a: Finance rejects – Credit Limit → auto-email Sales Org
- Exception 4b: Finance rejects – Other → auto-email original sender
- Exception 5: Finance approves → system auto-generates Excel and auto-emails Global ERP team (no human touchpoint)

SECTION C: Enterprise Security and Handover Mechanics

- Adaptive Card approval audit trail (Azure AD, timestamped, non-repudiable)
- Billing Template population from List data only – not from original email
- Operator dashboard: 5 Microsoft Lists views covering action queue, live pipeline, exceptions, completed today, and monthly reporting

Tone: Professional, authoritative, exhaustive. No placeholders.

HOW TO ADAPT THIS PROMPT

Replace These Elements for Your Industry or Use Case

Element	Replace With	Examples
Medical Devices	Your industry vertical	Pharma, FMCG, Auto Parts, Chemicals
Shared Outlook inbox	Your order intake channel	WhatsApp, EDI, Web portal, ERP
3 Hard Rules	Your compliance constraints	GST validation, credit limits, SKU MOQ
Credit limit → Sales Org routing	Your exception routing logic	Price deviation → Pricing team, Shortage → Planning
Microsoft 365 stack	Your enterprise tooling	Google Workspace, SAP, Salesforce
4 stakeholder personas	Your key decision-makers	COO, CFO, Regional Head, IT Lead
Power Automate	Your automation platform	n8n, Zapier, SAP BTP, UiPath

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